Gosford High School

INFORMATION SYSTEMS

An introduction to Year 9 Computing Technology



Year 9 Computing Technology

Welcome to Year 9 Computing Technology!

Over the next several lessons we're going to do a bit of a crash course in information systems. We'll cover the following areas:

- What are Information Systems?
- Hardware
- Software
- Data and Information
- People and Processes
- Issues and Cybersecurity

Whilst your teacher goes through slides with you, make sure to write down the definition of terms which your teacher indicates are important. At the back of this book, you will find space to do this and the list of terms you need to define.

You will then be given time each lesson to read over the notes in this booklet and answer questions.

It is understandable that you are likely keen to get stuck into more practical work, but first it is important that you empower yourself with the basics of computing so you know how it works.

Contents

3.1 WHAT ARE INFORMATION SYSTEMS?	5
ACTIVITIES: 3.1 INFORMATION SYSTEMS	11
3.2 HARDWARE	12
ACTIVITIES: 3.2 HARDWARE	16
3.3 SOFTWARE	17
ACTIVITIES: 3.3 SOFTWARE	20
3.4 DATA AND INFORMATION	21
ACTIVITIES: 3.4 DATA AND INFORMATION	26
3.5 PEOPLE AND PROCESSES	27
ACTIVITIES: 3.5 PEOPLE AND PROCESSES	30
3.6 ISSUES AND CYBERSECURITY	31
ACTIVITIES: 3.6 ISSUES AND CYBERSECURITY	35
GLOSSARY OF TERMS	36

IN ACTION

City2Surf information system

Every year, more than 80 000 people register to participate in the world's largest fun run, the Sydney City2Surf (Figure 3.1). Although this event is a huge feat of physical strength and endurance, the extensive data processing going on behind the scenes is just as vital for the event's success.

When a participant registers and pays for the City2Surf online through the website, an account is set up that stores key information about the person and their entry, including their name, date of birth, gender and the group they are racing in. A bib number is also generated, which is used to uniquely identify every participant (Figure 3.1). As the City2Surf is also a fundraising event, each participant has a fundraising web page set up to which people can be directed to donate money.

Finally, it is recommended participants download the City2Surf app to help them get 'race-ready' (Figure 3.2). This app includes much of the functionality on the website plus suggested training programs.

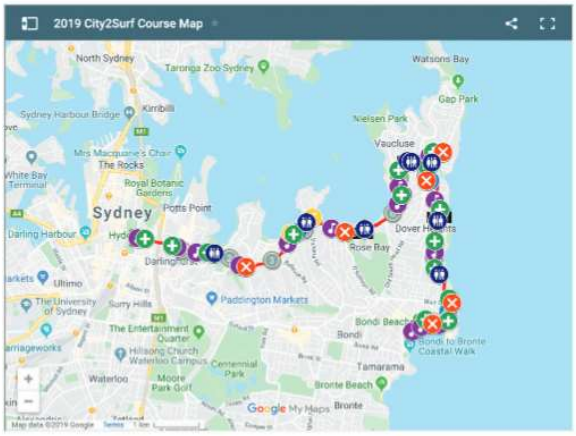
Before the event, participants receive or collect their race bibs, which include a special piece of technology that allows their race time to be accurately recorded. Embedded in the plastic bib is a radio-frequency identification (RFID) tag. At the start, end and at various checkpoints along the race route, the participants run over timing mats that contain antennas that receive radio signals from the tags. The sensitivity of the technology allows the system to keep track of all 80 000+ participants. These signals are then processed into time stamps, which are instantaneously transmitted to a centralised database and stored against each participant's record. During the event, photographers take photos of the participants. When the photos are uploaded to the City2Surf website, optical character recognition (OCR) software is used to determine the bib number in the photo and link it to an account.

After the event, participants are able to search for their results in the public access database or log into their accounts to see more detail about their results on their dashboard. Low-resolution versions of the action photos are linked to their accounts with the option for purchase of the full-quality images. SMS results are also sent to those who paid for that option at registration.



Getty Images/Brendon Thorne

Figure 3.1 All of the 80 000+ runners and walkers have registered for the event online and will have their results added to online accounts at the end of the race.



Map data ©2019 Google

Figure 3.2 Interactive course map showing the location and distance of key features along the route, including checkpoints, toilets, medical support, water stations and entertainment.

Questions

1. What is RFID? What other information systems use RFID technology?
2. Suggest three reasons why it is better to use an automatic tracking system rather than manually recording times.
3. Why would an app be more useful to a participant than the website on race day?
4. What is another technology the participants could use to keep track of their progress around the course?
5. What is OCR? What other information systems use OCR technology?

3.1 What are information systems?

In the most general sense, a system is a group of connected things that work together to achieve a goal or outcome.

Consider our respiratory system. It is made up of many organs and muscles in our body, including the mouth, nose, trachea, lungs, rib muscles and diaphragm. These all work together to allow us to respire, or breathe, providing oxygen to, and removing waste from, our blood. Our solar system is composed of the sun, planets, moons, asteroids, comets and meteorites, all moving based on the gravitational forces acting on them.

A digital system includes **hardware** and **software** components integrated to perform a multitude of possible functions. Many digital systems today also involve **networks**, such as the internet, allowing the functionality to occur across many locations and sometimes huge physical distances. A digital system allows you to video chat with a friend on the other side of the world or control the temperature of your house from your mobile phone.

When we consider a digital system as including **data**, people and processes, it is referred to as an **information system** (Figure 3.3).

Purpose of an information system

As the name suggests, the purpose of an information system is to produce **information**, commonly to solve a problem or identified need.

In this course, you will be developing a range of information systems, however, there will be an emphasis on **enterprise information systems** in early chapters. You will be applying knowledge and skills on how to use different hardware and software with people and processes in order to transform collected data into useful information.

Components of an information system

Hardware

Hardware is the physical components of an information system. A typical information system will include:

- one or more computers
- external or peripheral input and output devices, such as mice, cameras, printers and speakers
- networking infrastructure, such as a server, router and network cables.

Consider the City2Surf information system outlined at the beginning of the chapter. The hardware components of this system include the RFID tags and scanners, the computers used by the people who run the system to check on the data, the server used to store all the results and to host the website, and all the networking cabling and hardware used to connect these components together. Other hardware components of this system include computers used by participants to access the website and mobile devices used with the app. Cameras also form an important part of this system. Even the laser cutter used to engrave individual nameplates that fit into medals is a component of the system.

Software

Software is the instructions that allow hardware to process data. These instructions are initially written in one of many programming languages then transformed into raw binary instructions for the computer, called machine code. The software you would be most familiar with is the application software that allows computers, generally controlled by **users**, to perform a wide range

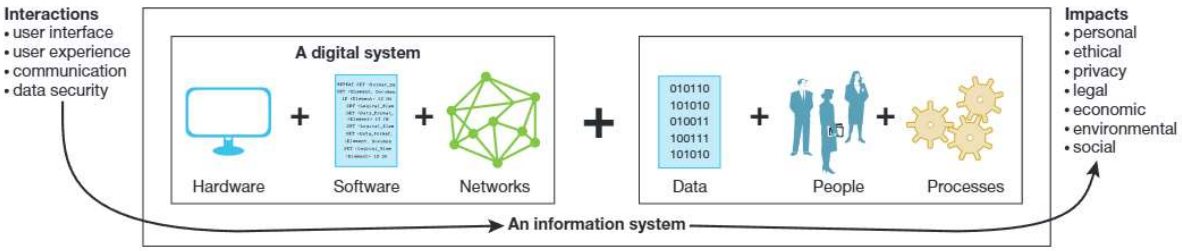


Figure 3.3 Information systems are created when a digital system becomes part of a larger system, along with data, people and processes.

of processing tasks, such as word processing, image editing and animation creation. Other software that goes on behind the scenes includes operating systems and utility software.

In the City2Surf information system described earlier, all computers would be using operating systems and utility software for the back-end control of the hardware. Many software applications are also involved, including the internet browser used to access the website, the database management software used to store and process user accounts and results, and the mobile application (Figure 3.4).

Networks

Networks are formed when two or more computers are connected together to allow the sharing of data. Connections are made using physical media, such as ethernet cables and optical fibre, or wireless technology, such as Bluetooth and Wi-Fi. Many networks rely on the biggest network of all, the internet, while other simpler networks allow more direct transfer of data, such as the use of Bluetooth to upload data from a smartwatch to your phone, or sending a print job to your printer on your home Wi-Fi network. There are lots of specialised networking hardware and software involved in establishing a fast, secure and reliable network for an information system.

In the City2Surf information system, each checkpoint would have included one or more computers that were all connected to a centralised server computer where data was transmitted for processing. The internet is also a key feature in this system for users, allowing them to create accounts, use the app and collect results (Figure 3.4).

Data and information

Data is the raw facts collected, stored and processed by an information system. When data is processed in a way that gives it meaning for humans, it is called information. Data is collected, stored and processed, and the resulting information is displayed by the hardware and software of an information system.

Consider again the City2Surf information system. The raw data collected by this system via the website includes name, date of birth, gender and address. One of the first processes performed on this data is to determine the category of participant defined by their age (calculated from their date of birth) and gender. During the race, the raw data collected from the RFID tags is timestamps at each location checkpoint. The data is processed to determine the final and section times, as well as the runner's pace. This data creates information that is further processed, along with other data, to determine total times for teams and ranks for participants and teams.

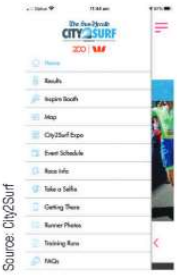
People and processes

People include the system's users and those involved in the operation and maintenance of the system. Although advances in artificial intelligence (AI)



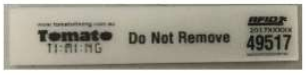
Source: https://city2surf.com.au/

The City2Surf website contains all important information about the event and allows people to register and check results.



Source: City2Surf

The City2Surf mobile application supports other functionality including things you would need while participating in the event.



The running bib includes an RFID timing chip, which can transmit the bib number to any receiving antennae within range.



The antenna in the timing mats collect data from the running bib RFID tags as people run over them. The time, combined with the runner's ID, is transmitted and stored in a database.

Photo by Jeffrey Chan © Canberra Times/ACM



Shutterstock.com/EML

A DSLR camera is commonly used for action photography. The digital images are stored on an SD card inside the camera, which can then be inserted into a computer to upload the photos. Many DSLR cameras these days also allow for direct wireless transmission of photos from the camera to a computer.

Shutterstock.com/monan11



OCR software identifies bib numbers within the image and determines all the digits using AI algorithms based on machine learning. This calculated bib number allows the photos to be linked to entrants' accounts.



Source: Intellipaat

A server is a large computer used for storing data. Companies may own and manage their own servers or use a service like Amazon Web Services to store data 'on the cloud'. These services also provide the software for the management of the data, such as a database management software.

Shutterstock.com/Gorodenkoff

Figure 3.4 Software, hardware, networks, data, people and processes are all involved in the City2Surf information system.

are supplanting many tasks previously requiring human involvement, many systems still require the participation of people, with the success of the system commonly relying on their roles and responsibilities.

Processes are the steps and rules followed to allow for an information system to perform successfully. It may be helpful to think: 'Processes are to people what software is to hardware'.

In the City2Surf information system, there are many people interacting with the system at different levels of control. Apart from the people participating in the event, there is a person in control of starting the official clock at the start of each wave, which will be used to determine final times, and the photographers who take photos and upload them to the centralised storage platform. There are also numerous people involved with checking and maintaining the system to ensure any issues or errors are corrected.

Structured vs unstructured data

Data that we would like to process into information may be structured or unstructured, impacting the information systems and processes we use to collect, store and analyse the data.

Structured data

Structured data is generally quantitative data that fits into predefined formats or models. It is commonly stored in databases with records for each entity and event, and fields for the attributes of these entities and events. For example, in the City2Surf, the structured data includes all the attributes for the participant entities including their names, dates of birth and bib numbers as well as the attributes for the events that occur as they run the race such as passing a checkpoint (time attribute). When data is structured, it is easier to store and analyse.

Unstructured data

Unstructured data does not fit into predefined models and can be very complex. Photographs are a common example of unstructured data. In the City2Surf, the photographic data are able to be analysed but require other technology such as OCR. The analysis of unstructured data may require machine learning.

Three Vs of data

Data should be evaluated before processing to ensure the most efficient and accurate methods are used. Considering the three Vs of data is one way to do this. Variety refers to how structured, unstructured or semi-structured the data is as described above. The second V is volume and refers to how much data we have and from how many different sources it is aquired. Velocity is the final V, referring to the speed at which the data is generated.

Evolution of data analysis

Advances in AI have allowed for analysis of unstructured data and the links between previously siloed collections of structured data. Machine learning does not require predefined models, instead it looks for patterns, some not seen before, to bring structure and meaning to the data. In 2020, the City2Surf organisers wanted to analyse all of the past results to determine some key information about participants over the years including demographics and returning participants. Early years' data was in the form of scanned newspapers. AI algorithms were used to extract names and race times from these images, which were then stored with other data in a cloud-based data repository. Other machine learning algorithms were then used to find relationships between all the data and to provide key information such as the City2Surf 'legends', who had participated in all 49 events since it first began in 1971.

INFOBIT

Pokemon Go is a game that relies on a range of modern **data analysis** techniques for structured and unstructured data. In this game, players use their smartphone cameras to capture images of real-world locations, which are then analysed by a classifier to identify and overlay virtual Pokemon on the camera view. The classifier uses data analysis and machine learning techniques to identify features in the captured images including terrain and environments that correspond to specific types of Pokemon. For example, water-type Pokemon are more likely to appear near bodies of water, while grass-type Pokemon are more likely to appear in parks and other natural environments. This involves analysing the colour, shape and texture of the images, as well as identifying landmarks and other features that can be used to pinpoint the location of the virtual Pokemon. The image recognition system used in *Pokemon Go* is a good example of how classifiers can be used in video games to create immersive and interactive experiences.

Categories of current information systems

There are numerous types of information systems, some with very specific purposes and others designed to meet a range of needs. Following are some systems widely used today in organisations.

Management information systems

Management information systems involve the processing of data for organisations. They aim to increase an organisation's value by improving efficiency through coordination, control and automation of data. Analysis and visualisation of data helps support

decision making. Management information systems commonly include a range of other information systems, such as transaction processing systems.

Transaction processing systems

Whenever a purchase is made at a register or online, a transaction processing system is at work. In general, a transaction is whenever an exchange occurs or business is done between two parties. This mostly involves the buying and selling of products, but also relates to the payment of employees and when business occurs that does not necessarily involve money, such as a reservation or a loan. Transaction processing systems collect, process and store data about transactions. If an error occurs at any stage during the processing, such as a buyer having insufficient funds in their account, the transaction fails. Data can be processed in real time (for example, when you make a contactless payment using a debit card) or it can be done in batches, as in the payment of employees.

Decision support systems

Decision support systems provide guidance for individuals and organisations attempting to make a decision about a situation that cannot be clearly defined by known variables. These situations are called unstructured or semi-structured. For example, if we were deciding on what type of dog to get as a family pet, there are some clear variables that can help in the decision process, such as the size of yard and time to spend on grooming. However, there are many other more subjective or little-known variables that could also be useful to consider. A decision support system built using a knowledge base of all the possible variables, their weightings of importance, interactions and known outcomes could be used to help you make your decision of the most appropriate dog type. It is important to remember that decision support systems just *support* the decision process. In the end, it is the user that makes the final decision.

Expert systems

Expert systems were one of the first forms of AI. Their purpose is to replace the role of an expert in a particular field, allowing users to query the system to find answers to questions or to find evidence to support their theories.

An expert system is based on a knowledge base of facts and rules, which are defined using knowledge from an expert, and an inference engine that processes the queries from the user using the knowledge base to form a conclusion.

Expert systems are commonly used in identification and decision making (Figure 3.5). However, they are being increasingly replaced with AI identification systems that use neural networks to learn and improve each use.

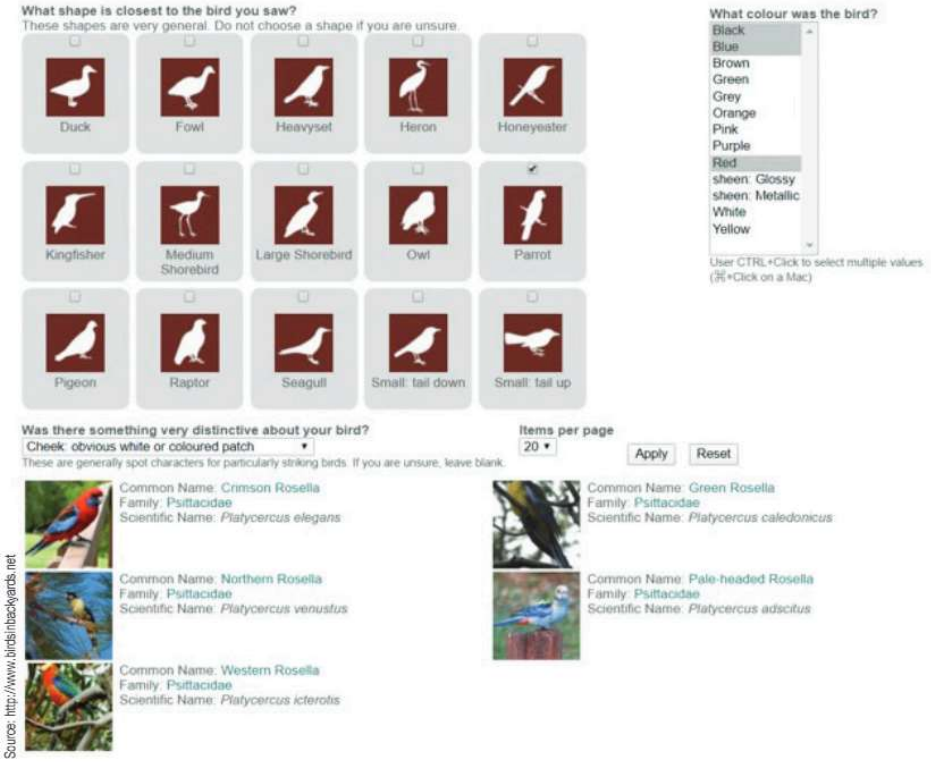


Figure 3.5 The website Birds in Backyards' finder application is an example of an expert system. This expert system allows a user to identify a bird they may have seen in their backyard based on observable features such as size, shape and colours.

Geographic information systems

Geographic information systems (GIS) layer many data sets over one another on a geographic plane. This provides location-based analyses and visualisation, which can aid in communication of data and reveal hidden trends and patterns. Google Maps is a comprehensive GIS that includes data on streets, businesses and satellite imagery as well as real-time data on traffic, speed cameras and location (Figure 3.6).

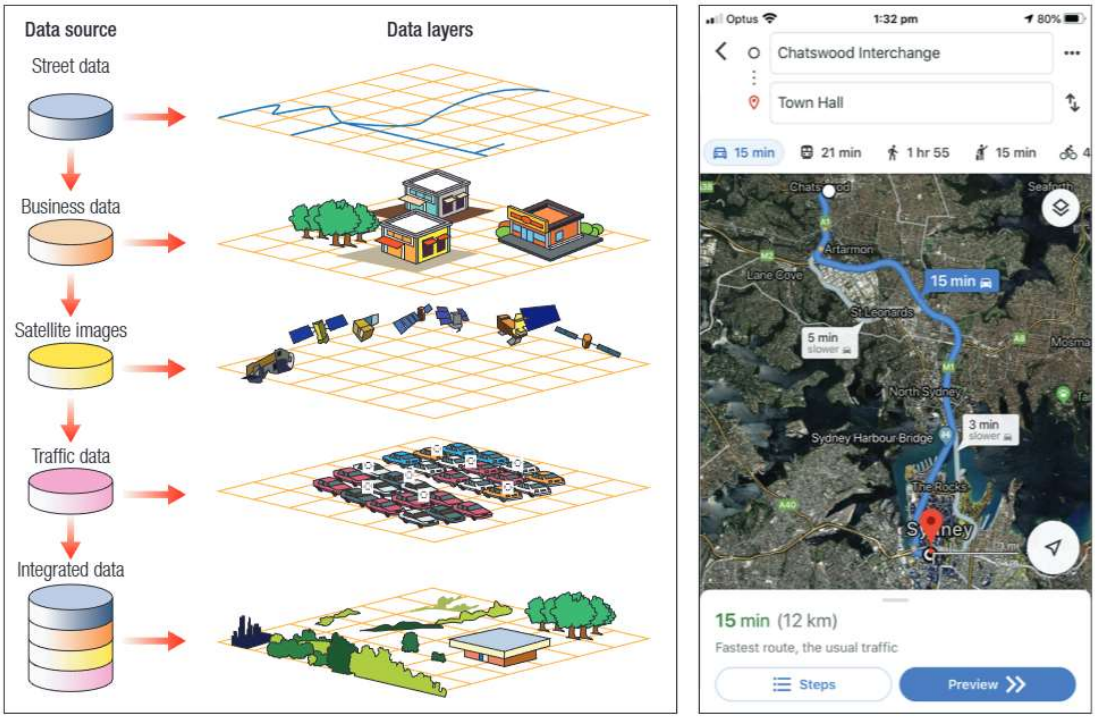


Figure 3.6 Some of the different data sources integrated into Google Maps.

Enterprise information systems

Enterprise information systems are the ultimate management information systems and are used in organisations such as businesses, government, schools, charities and clubs (Figure 3.7).

The functions of an enterprise package could include online payment processing and automated billing systems, security, content management, customer relationship management, business intelligence, project management, collaboration and human resource management.

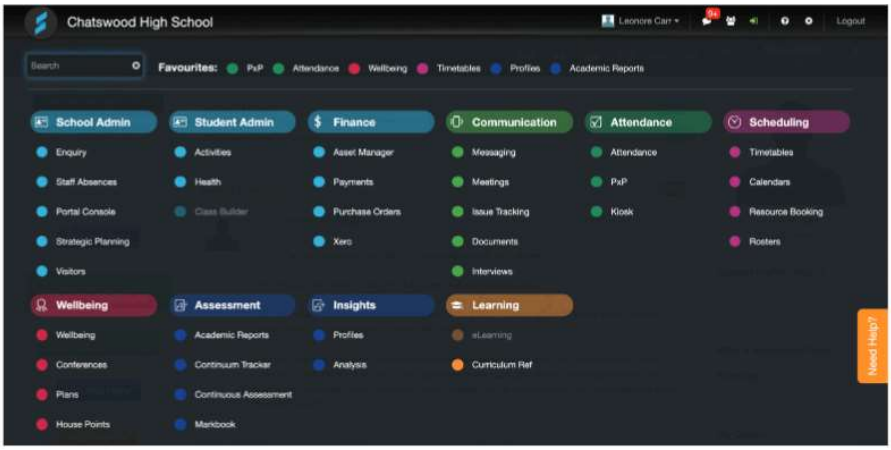


Figure 3.7 Many schools use enterprise systems such as this to allow them to manage attendance, reports, timetables, student payments, wellbeing documentation, staff communication and interview scheduling.

The early chapters will introduce you to technologies and processes similar to those used in enterprise information systems. Modelling the same principles of enterprise information systems within the context of your smaller scale experiences will help in your knowledge and skill development.

Learning management systems

A learning management system (LMS) is designed to manage and deliver educational courses, training programs, or learning and development activities. LMSs provide an online platform where instructors can create, manage and deliver learning materials, track learners' progress and performance, and facilitate communication between learners and educators. Some popular examples of LMSs include Canvas and Google Classroom.

Multimedia systems

A multimedia system allows users to interact with and manipulate multimedia content, which can include a combination of text, graphics, audio, video and animations. Multimedia systems are designed to provide an immersive and engaging user experience by presenting information in a variety of formats, and can be used for a wide range of purposes, including entertainment, education, communication and business. Multimedia systems typically consist of a combination of hardware and software components, such as a computer, a monitor or display, input devices such as a keyboard, mouse or touch screen, and multimedia authoring tools or applications. Examples of multimedia systems include video game consoles, streaming media players and interactive kiosks.

INFOBIT

Keeping Culture is an online platform designed to store and safeguard Aboriginal and Torres Strait Islander Peoples cultural knowledge and media. The system organises the media, including photos, documents, movies and sounds, into archive records, and the knowledge into profile records, such as people and places. These are all linked to create a network of knowledge. This system is called a knowledge management system and incorporates elements of geographic information, enterprise information and multimedia systems. To find out more, go to the Keeping Culture website.

Tools for information systems

There are a number of common tools or key features integrated into many current information systems. You will learn about these in more detail in later chapters. In sections 3.2 and 3.3 a range of hardware and software and their functions in different information systems are introduced.

Databases and data warehouses

Organisations collect data from a wide range of sources relating to their operations and management. This data is organised using databases and processed using database management systems. By aggregating these databases and creating data warehouses, organisations can use specialised tools and processes such as data mining to gain useful, possibly hidden, information to guide future management decisions.

Combining separate databases can raise significant ethical issues, as users are generally unaware of the new information that can now be deduced about them.

Spreadsheet software

Quantitative data analysis is commonly built into commercial enterprise software. However, organisations or individuals may see the need to develop their own custom software systems for a particular purpose. Spreadsheet software, such as Microsoft Excel (Figure 3.8), can form the basis of many of these internally developed systems for data analysis, such as stock tracking and budgeting. Spreadsheet software can also be used to develop database management systems when the organisation of data stored is not overly complex.



Shutterstock.com/Pawpixel.com

Figure 3.8 Microsoft Excel is the most widely used spreadsheet software, in industry, education and for personal use. Google Sheets is a strong competitor to Microsoft Excel as it has the advantage of being freely available, a straightforward interface and is web-based. Both offer very similar features, so users can easily transition between them.

Weblink
Keeping culture

Artificial neural networks

For true automated insight that operates and grows along with the organisation, an artificial neural network can be implemented as part of an information system.

Artificial neural networks are programs that learn. They provide many data analytics and decision-support functions much like other information systems; however, they also involve machine learning to improve their performance with each experience. For example, an artificial neural network may be used in an automated online chat system on a business's website. After each exchange that a chatbot has with a customer, it will learn what terms and phrases are associated with the services the customer is seeking.

ACTIVITIES: 3.1 INFORMATION SYSTEMS

1. Identify the components of a digital system.

.....

.....

.....

.....

.....

2. What is the difference between a digital system and an information system?

.....

.....

.....

.....

.....

3. Explain the phrase 'Processes are to people what software is to hardware'.

.....

.....

.....

.....

.....

.....

.....

.....

3.2 Hardware

A computer can be defined as an **information processor** – we supply it with data to be changed into an appropriate format, which can then be stored or displayed for a specific purpose. Hardware includes a range of physical devices – items that you can see and touch. Figure 3.9 shows a picture of a typical computer hardware system. The hardware components work together to complete the overall task.

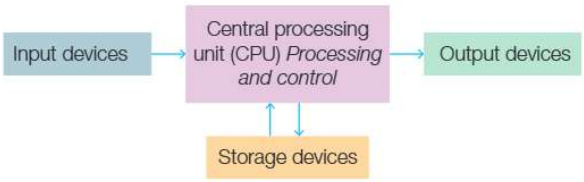


Figure 3.9 Hardware components of a computer system and the data flow between them. Most of the devices connected to the computer are under the control of the CPU. Data and information flow into, around and out of the system.

Hardware functions

The components of a computer system such as the one in Figure 3.10, are classified according to the type of function they perform – input, output, storage, processing and control. The hardware of a computer system has a part to play in all of these functions.

Input

An input device collects data from the outside world and changes it into a form that the computer can work with. The keyboard, mouse and touchpad are probably the most commonly used input devices. The keyboard will convert the press of a key into an electrical signal to be transmitted either wirelessly or along a cable to the system box. Different signals will indicate which key has been pressed. This data is stored in memory until it can be processed by the central processing unit **(CPU)**.

Microphones, scanners, digital cameras and graphics tablets are also common input devices. Some systems have sensors that collect data such as temperature, light, noise and movement that are sent to a central unit for processing into appropriate information.

Output

An output device communicates information from the computer to the user. Information may be output as a display on a monitor or printer, or it may be output as sounds from a speaker. The monitor converts digital electrical signals from the system into patterns of light on the screen, which can be interpreted by the human brain as text or images. Head-up displays, 3D printers and laser cutters can also function as output devices. Speakers take electrical signals output from a computer and convert them into sound waves, perhaps speech, music or sound effects.

Storage

There are two types of storage in a computer system: primary and secondary.

Primary storage is short-term, *volatile* memory, meaning it is only stored while the computer is running. Two main examples of primary storage are random-access memory (RAM) and cache.

There are three main technologies associated with secondary storage: magnetic, optical and solid state (or flash) memory. Not that long ago it was common for people to use portable hard disk drives (magnetic)

to back up their computers, DVDs or CDs (optical) to share relatively small collections of files. This has now been largely superseded by flash memory alternatives, including flash hard drives and USB sticks, which offer large storage capacity, fast transfer speeds and good portability, all with ever-reducing costs.

You will already understand the need to save your files for future use. Storage devices can be local or remote. Local external drives are being replaced in many cases by online storage using remote servers, known as cloud storage. Networked and online file servers are designed to allow secure storage of data and are generally accessible across the internet. This reduces the risks involved in physically carrying around portable devices, such as accidental loss or damage, and also offers the advantage that the data may be accessed by many people from various locations.

Processing and control

The CPU is widely referred to as the brain of the computer. It can comprise a single integrated circuit (chip) or a number of chips working in parallel. It is responsible for the processing and control of all the data flowing into, out of and around the system.

Computer chips are made up of millions of tiny electronic components called transistors. A transistor is really just a switch. By combining these switches into complex circuits, **microprocessors** are able to perform a variety of complex tasks. Some of these chips (for example, RAM chips) are classed as storage devices, as they store programs and data that need to be instantly accessible to the CPU. Other chips act as processors to carry out routine tasks, such as preparing audio and video data for output.

RAM is volatile, which means it loses all its data when the power is switched off. That is why you need to save your files to secondary storage devices such as your hard disk. Read-only memory (ROM) chips hold information that the CPU needs on a regular basis, such as when the computer is starting up. This data cannot be changed by the user. These ROM chips hold their content even after the power is switched off, otherwise your computer would not be able to start or perform other essential operations.

Graphics processing unit (GPU) chips were designed primarily to manage and boost the performance of video and graphics processing tasks. While the CPU handles most of the general computational workload, the GPU takes care of graphics-intensive tasks, such as rendering videos, playing video games and running graphical applications. In recent years, GPUs have found use beyond graphics processing. Their ability to handle parallel tasks efficiently has made them useful for other data-intensive tasks, such as cryptocurrency mining, scientific computing and artificial intelligence applications.



Figure 3.10 Most parts of the computer system perform a specific task. These tasks can be classified as input, output, storage, processing and control. Can you identify and classify each hardware item illustrated?

Classification of hardware systems

Computers are classified according to how powerful they are, and this generally determines what type of job they do. System classification is used to give people a general idea of what the hardware can do. These classifications are generally a little blurred because technology changes so rapidly. For example, mobile devices such as phones and tablets now have the technical power and functionality that only desktop personal computers had previously. Figure 3.11 shows examples of hardware systems and Figure 3.12 shows a comparison between them in terms of time, cost and processing power.

Microprocessors

Microprocessors are tiny electrical circuits made up of millions of transistor switches, mounted on a silicon chip, that control the flow of electrical signals. Much of what a computer does is dependent on these electronic switches routing data in different ways, according to the software instructions. The CPU inside the system

box of a personal computer is commonly referred to as the processor, or microprocessor, as it is very small and controls the main operations of data processing.

As well as computers, microprocessors are found in many different electronic devices. Their job is to store and process data – that is, to change data into a form that is useful to humans or another device. These devices include cameras, watches and monitoring devices.

Smartphones

Smartphones provide many more functions than just phone services, and their wide use and acceptance are evidence of their extensive functionality. As well as being portable, they also boast many more sensors than are found in personal computers.

Most of their data processing is done on the servers they connect to, although the high speed at which they transmit and receive data tends to obscure this.

Personal computers

A personal computer is any computer that can perform all of its input, processing, output and storage activities by itself. It is small enough to sit on a desktop, although the system box may be positioned on the floor. A workstation is a networked desktop computer that can be used in combination with a server or other computers as part of a network for a wide range of tasks, including complex calculations and graphics processing. A notebook or laptop computer is a light and portable personal computer, yet still powerful enough to meet most needs of the individual worker.

Servers

A server can be any networked computer located anywhere that has appropriate server software installed.

In larger organisations, servers are generally dedicated machines that control access to software, files and printers. They may be required to support thousands of connected users at one time. These host computers can be used on the internet to act as web servers.

Mainframe computers

Mainframe computers can cost several millions of dollars. They can store vast amounts of data and are accessed with terminals or personal computers. Banks and large government organisations use mainframe computers to process data. Users communicate with these computers using terminals.

Supercomputers

A supercomputer is the fastest and most expensive of all computers. Supercomputers will contain thousands of processors that can share jobs and work in parallel. This means that they are capable of processing many trillions of instructions in one second and can be used for complex mathematical calculations such as weather forecasting or scientific research. A supercomputer might cover the area of two basketball courts. They are used to make weather predictions, making sense of huge amounts of data.



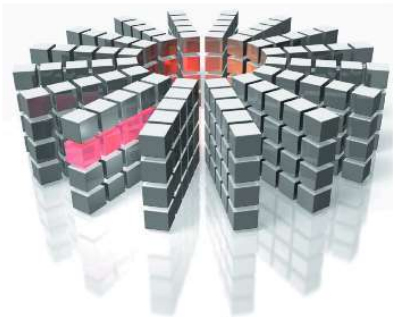
Shutterstock.com/Kaspars Grinvalds

A smartphone is now more like a computer than a phone.



Shutterstock.com/Den Rozhnovsky

A personal computer might be small enough to sit on your lap, yet powerful enough to act as a server on a network.



Shutterstock.com/cyclone project

Supercomputers are used for weather forecasting and scientific research.



Shutterstock.com/dotshock

Network servers are housed in cabinets and are easily removed and replaced.



Shutterstock.com/Timofeev Vladimir

Mainframe computers are used by banks to process millions of transactions per day.

Figure 3.11 Portability and power are two factors that must be considered when designing new hardware systems.

9780170442787

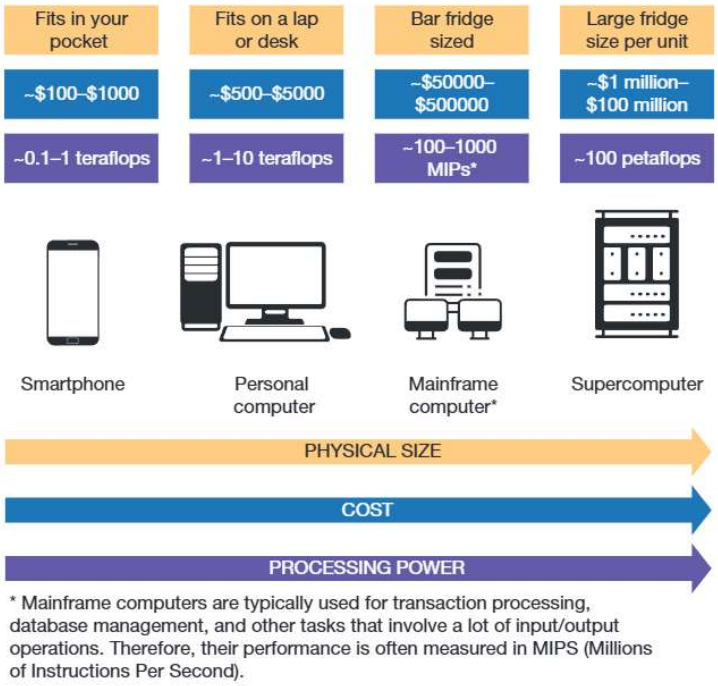


Figure 3.12 Not unexpectedly, as processing power goes up, so does physical size and cost.

Quantum computers

Unlike traditional computers that use bits to represent either 0 or 1, quantum computers harness the unique properties of quantum mechanics and use quantum bits or qubits. These qubits can exist in multiple states simultaneously, allowing quantum computers to perform incredibly complex calculations at speeds that are unimaginable for classical computers. Quantum computers built by IBM are already tackling problems that were previously beyond the reach of conventional technology, including complex simulations, supply chain optimisation and cryptography. AI is also bound to be significantly boosted by quantum computing.

Buying a personal quantum computer is a while off, but already people in industry can lease access to use them remotely.

INFOBIT

One of the earliest supercomputers, the CRAY-2 (1985), weighed ~2500 kg and had a processing power of 1.9 Gigaflops. The iPhone 14 Pro Max (2022) weighs only 240g and can hit 2 teraflops. That's 1000 times more processing in a system more than 10 000 times smaller.

ACTIVITIES: 3.2 HARDWARE

1. Draw up a table of hardware devices that you know of. Identify each one as a device for input, output or storage (do this by ticking the box for INPUT, OUTPUT, STORAGE). Three devices have been identified for

DEVICE	INPUT	OUTPUT	STORAGE
Mouse			
Speaker			
RAM			

2. Explain the difference between a microprocessor and a personal computer.

.....

.....

.....

.....

.....

.....

.....

.....

3.3 Software

Software systems enable hardware to do a wide variety of tasks. The software is a list of instructions that enables documents and data to be processed by the hardware. This software is created in a range of programming languages and can be written using graphical icons or English expressions.

Application software

Today's computers and digital devices have a range of uses, and each of them will need to have a particular type of software. Most people are familiar with application software such as word processors, graphics editors and web browsers that are able to perform a range of tasks in creating, editing, formatting and displaying documents (Figure 3.14). Increasingly, people are creating their own applications (apps) for mobile devices. These small applications can perform a single or limited task and can be created using open-source (free), user-friendly software.

Some devices, such as automatic teller machines (ATMs), routers and traffic-light controllers, perform just one or two, sometimes complex, tasks. Other hardware, such as a microwave oven, has software embedded in it so the devices are fast and reliable.

In general, application software is designed to allow the user to process data – for example, to create and edit a file, or to display and transmit information across a network. Consider the following categories of application software.

Communication software

Due to the ever-increasing accessibility of fast internet, people can easily communicate using a range of systems.

- Internet browser software is used to download a wide variety of internet and intranet documents. Browsers also enable users to interact with web applications – for example, online banking and shopping.
- Instant messaging software enables real-time communication between two or more people.
- File transfer programs allow users to upload files over the internet – for example, to web servers and wikis.
- Increasing numbers of companies are moving to internet protocol (IP) telephony or voice over internet protocol (VoIP), which enables workers to make lower-cost phone calls using their computer. Skype™ is a popular example of this.
- Videoconferencing software enables video and audio communication between users over the internet. These programs can provide other tools, such as chat and multi-user document editing.

INFOBIT

New Zealand is a geologically active country with regularly occurring earthquakes and risks of other natural disasters such as volcanic eruptions and tsunamis. New Zealand uses an Emergency Mobile Alert system that broadcasts warning messages to people's mobile phones within range of targeted mobile towers when there is serious threat to safety.

How does Emergency Mobile Alert work?



1. Authorised agency determines affected area and creates alert.
2. Alert is broadcast from relevant cell towers.
3. Capable mobile phones in the affected area receive the alert.

Source: National Emergency Management Agency (NZ)

Figure 3.13 A diagram from a fact sheet detailing the new mobile communication system that can alert people of an emergency in New Zealand

Business software

Business organisations depend on a variety of software tools to manage their workload effectively. Word processing and email software are essential elements in today's modern business. In addition, intranets or knowledge management systems enable information to be posted for reference by all or specific employees.

Project management software is useful for scheduling tasks and resources in large team projects. Resource planning software is used to manage resources such as staff and finances. This helps to keep all branches of a business fully informed of essential business information, especially in large companies that may be developing rapidly or located in different parts of the world.

Businesses use computer models to help them make decisions – for example, by identifying market trends. These can range from a simple spreadsheet to a powerful analytical tool custom-built by software specialists.

Software for schools

All students in New South Wales are required to meet specific standards when using computing technology. Consequently, most schools will have a list of standard software installed on their computers for students to use. Students who choose computing technology as an elective will learn to use more sophisticated software and probably in a more innovative manner than their peers. In addition, school administrators use software to create and publish reports, and to store and submit data to educational authorities. Most of this work is database-driven.

Science and research

Most areas of science and research have their own specially developed software to analyse large volumes of information and help them in their quest for new knowledge. Expert systems and AI are used and have potential for further development. Current applications include fingerprint matching by forensics teams and diagnosis of cancer cell malignancies.

Applications for mobile devices

Software applications on mobile devices are commonly referred to as 'apps'. People develop games and other apps for smartphones and tablet computers. User-friendly software is available to help with this, so that you do not have to be a specialist **programmer** to develop your own apps.

Operating systems

On a personal computer, an operating system is needed to work with application software to control the computer's hardware. Microsoft® Windows® and macOS are the most widely used operating systems around the world.

The operating system is the first thing we see on the screen when we start up a computer. It provides the interface that asks us to log on or it can display the desktop and its icons immediately. Similarly, when we close down the computer, it tells us that it is logging off and closes all the files in a systematic order before it actually powers down the hardware.

System requirements

When we want to install new software on our computers, we will probably be notified of the system requirements – that is, the minimum requirements in terms of hardware and operating system versions for the software to work properly (Table 3.1). Currently, manufacturers are producing 64-bit computers, which means the CPU, operating system and the application all have to be capable of processing 8 bytes of data in one instruction or data item.

The CPU

If a software package defines a 3 gigahertz (GHz) CPU and 4 gigabytes (GB) of RAM as its minimum hardware requirements, the package may fail to run properly on anything less than this, or it may run so slowly that you would be disappointed with the performance.

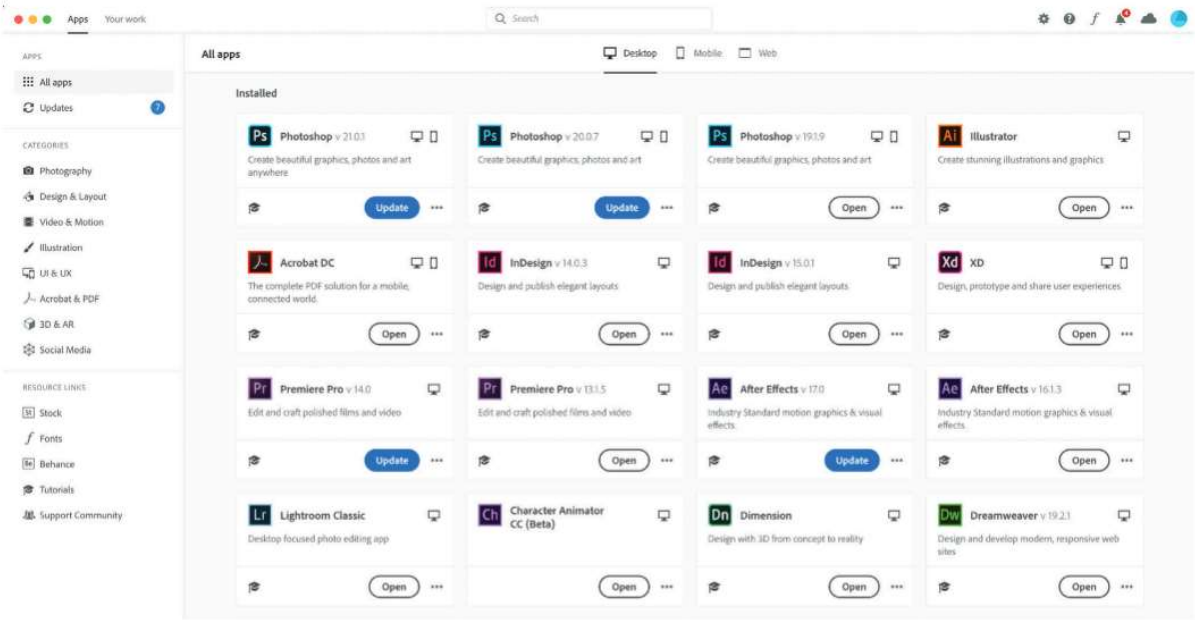


Figure 3.14 Adobe® Creative Cloud Suite® is a market-leading collection of applications available on desktop, mobile and web-based platforms.

Some programs use the CPU much more than other programs. Graphics programs that manipulate photographs and computer-aided design (CAD) systems generally use the CPU far more than word processing programs, because of the very large number and complexity of the calculations required to manipulate graphics. Specialised graphics cards containing a GPU are required for efficient operation of such applications. Businesses that need to work with the latest versions of software will need to regularly upgrade their hardware, as well as the software.

Demands on memory

RAM in the computer is used up by:

- the operating system of the computer,
- any programs currently running, and
- the data those programs use.

If a program uses a lot of memory to run, and it works with large files in memory, a computer system needs to have more than the minimum memory required by the operating system or a single program.

When programs cannot find enough memory, the operating system may use part of the hard disk to store some of the data in 'swap files', and this will slow down processing. This is referred to as virtual memory, where the computer looks at RAM for areas that have not been used recently and copies them onto the hard disk. This frees up space in RAM to work with the currently used applications.

Table 3.1 It is very important to check the system requirements for software before installing them. If the computer cannot meet the requirements, the application may not work as expected or at all.

Minimum system requirements for Adobe After Effects (Windows)	
Processor	Intel or AMD quad-core processor
Operating system	Microsoft Windows 10 (64 bit) versions V20H2 or later
RAM	16 GB minimum (32 GB recommended)
GPU	2 GB of GPU VRAM
Hard disk space	15 GB of available hard disk space; additional free space required during installation (cannot install on removable flash storage drives) Additional disk space for disk cache (10 GB recommended)
Monitor resolution	1920 × 1080 or greater display resolution
Internet	Internet connection and registration are necessary for required software activation, validation of subscriptions, and access to online services

Source: Adobe, 'After Effects (2023) system requirements'

Monitors

Size, resolution and speed of your monitor can impact significantly on the effectiveness of your application. Monitor size should be decided on before you buy your computer system. Resolution and colour depth can be set in the properties window of your display. Computers used for gaming and other processes with a highly dynamic display need monitors with a high refresh rate, such as 144 Hz, which means the image refreshes 144 times per second.

Storage

Secondary storage is essential as a non-volatile means of saving your documents. It also stores the software applications and the operating system, and loads a copy of them into RAM as required. However, as software applications become more powerful and sophisticated, they will become larger and so require more space on the hard drive. Similarly, so will the documents they create. Many people use online storage such as that provided by Dropbox or Google Drive instead of purchasing their own external drives, although online storage may involve an annual fee.

ACTIVITIES

Identify

1 List application software you use frequently.

2 Identify each category of application software and provide an example of each.

3 When installing new software on a computer, what system requirements need to be considered?

Analyse

4 A company opts to use videoconferencing software for a meeting between employees in offices at different locations. What are the advantages and disadvantages of using videoconferencing instead of face-to-face meetings?

5 Open a range of application software and compare the system requirements. Summarise your findings in a table.

Extend

6 Interview an adult friend or family member about software they use in their place of work. Outline the processes involved in some key functionality of the software.

ACTIVITIES: 3.3 SOFTWARE

1. Identify each category of application software. Provide an example of each.

.....

.....

.....

.....

.....

.....

.....

.....

2. Compare application software and operating systems.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

3.4 Data and information

Data and information are all around us. As human beings, we collect, process and store information all the time. The functioning of our daily lives relies on our access to information, but our brains can only cope with so much. For this reason, most people and companies rely on digital technology to do the collecting, processing, storing and transferring for them. The difference between us and a computer is that we can make sense of what we hear and see even when it is presented in many different ways. Computers require data in digital form and they need to be told exactly what to do and how to do it.

In its simplest form, **digital data** is either a zero or a one – a **'bit'** of data. Computers store and perform calculations on this data in very large amounts, very quickly. This gives us access to huge volumes of information that we can process and present again in a variety of forms for a variety of purposes.

In this section, we consider how data is entered and stored by a computer system, and the different forms it has to take to allow the user to perform a particular task. We will look at where and how data is used to produce information and how this in turn can form data for further analysis.

Data versus information

Many people confuse the terms 'data' and 'information'. Imagine a list of numbers written on a piece of paper. This might be called data because it has no real meaning for anyone. However, if someone told you that the numbers were telephone numbers, then they would have some meaning. The numbers have now become information.

Think about a sentence typed into a computer from a keyboard. The individual letters of each word are entered by pressing the keys. The letters are items of data for the computer to process. A word processor is used to store the letters and to display them on the screen. The output will make sense to a person, but for the computer, the display is just an arrangement of coloured dots on the screen.

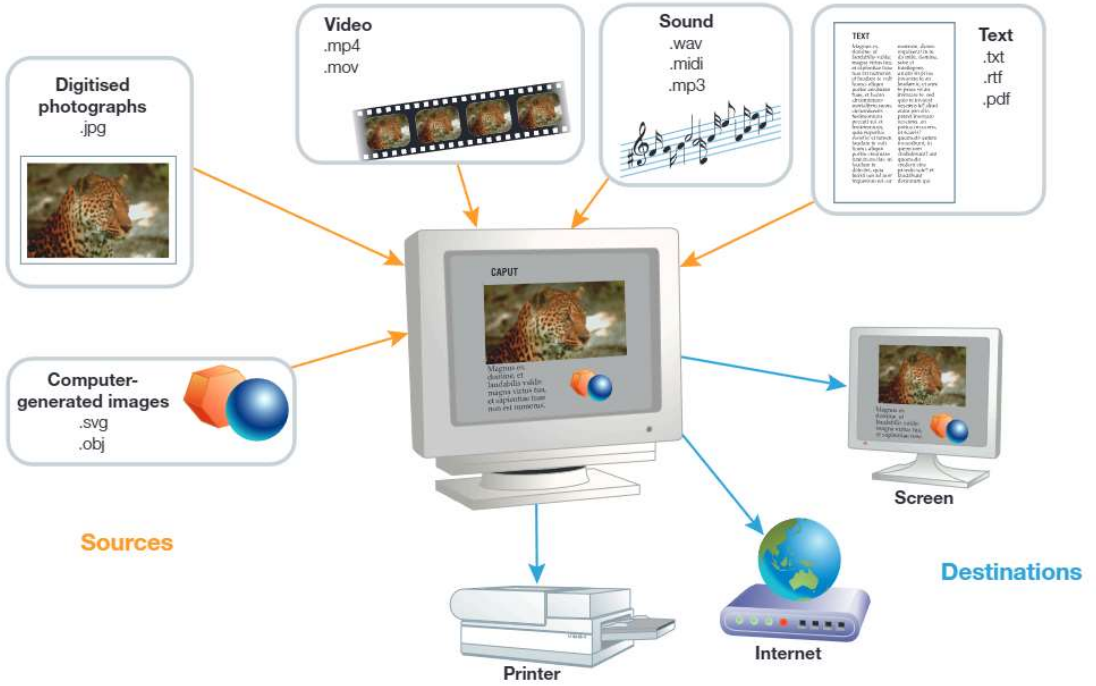


Figure 3.15 Information becomes data when we enter it into the computer for processing. It can come from a variety of sources and in many different forms, all represented using the binary system. This data becomes information again when it is displayed and interpreted by people.

As you work at the keyboard, the information in your head becomes data in the computer system. As you recognise the dots on the screen as words, so the data changes back into information.

Data can be letters or numbers, a pattern of dots, or a series of electrical or light signals. Data becomes information when it is given meaning by people.

Data forms

Most of the information around us comes in variable quantities – for example, the volume of music playing or the speed of a passing car. If the values of these quantities can vary along a smooth line or curve, we refer to it as **analog data**. Some information, however, comes only in fixed quantities – counted values, for example. This is called **digital data**. The values on a digital watch do not represent time passing in a smooth and continuous way, they only display the precise number of hours and minutes that have passed since midnight.

Digital data

Modern computers are designed to work with digital data. This means that all the data that enters the system has to be digital or has to be digitised before the computer can store or process it. Most input devices, such as a keyboard or scanner, are designed to provide the system with digital data. They input data to the system in digital form. However, a microphone inputs an analog electrical signal that is digitised by the sound card inside the computer.

Data coding

While a digital signal is allowed to take several distinct predetermined values, most computer systems work with switches or signals that can either be 'on' (1) or 'off' (0). Since the computer works with just these two values, we call them binary (meaning two) digits or bits. This arrangement allows computers to work very quickly and accurately.

We may find it difficult to work with binary numbers, but with a little practice it can be fun. We can use a series of binary digits, or bits, to represent bigger numbers and many patterns of data. The computer's work is performed using streams of zeros and ones (Figure 3.16).

So what does the series of bits 01100110 mean to a computer? It depends on how these bits are being used. It is very important to understand this clearly. If it is a signal coming from a keyboard, it might represent a character. If it is going to a monitor, it might be the colour of one of the pixels, or it might be part of a program instruction in the CPU. Remember that each bit represents a unit of data, so a series of eight bits, or a byte, can represent 256 or 28 patterns of data, including zero. That is enough for 256 colours on the screen or characters on the keyboard.

Counting in decimal

In the decimal system, each column value is 10 times more than the one on its right. We count up to 9, then we have to put a 1 in the tens column, and so on.

Hundreds	Tens	Ones
		0
		1
		2
		3
		4
		5
		6
		7
		8
		and so on.

Counting in binary

In binary counting, each column value is two times more than the one on its right. We count 0, then 1, then we have to put a 1 in the twos column, and so on.

Eights	Fours	Twos	Ones
			0
			1
		1	0
		1	1
	1	0	0
	1	0	1
	1	1	0
	1	1	1
1	0	0	0
			and so on.

To convert the binary number 1101 to decimal, we just read off the column values where there's a 1, and add the decimal values together.

	Eights	Fours	Twos	Ones	
Binary	1	1	0	1	
Decimal	8	+ 4	+ 0	+ 1	= 13

To convert back to binary from decimal, just share out the biggest one. For example, 43 in decimal is one lot of 32 plus one lot of 8 plus one lot of 2 plus 1.

32s	16s	8s	4s	2s	1s
1	0	1	0	1	1

Figure 3.16 Counting in the decimal and binary systems.

INFOBIT

Analog computers were virtually completely superseded by the more precise and repeatable processing of digital computers. In recent years, with advances in AI and quantum computing, analog computers may be making a come back.

Data types

It is amazing to think that the almost perfect recording of your favourite band or the beautiful rendering within the 3D world of a game you are playing is defined simply by zeros and ones. But how does the computer know how to distinguish one set of binary data as a 3D animation and others as audio, graphics, text and so on? A file format defines how these zeros and ones are organised and coded so they can be interpreted for the intended display. For example, the sequence of bits 01001101 could be interpreted as: the letter 'M', the decimal number 77, the red value in the RGB code for a pixel of colour 'cornflower blue', the amplitude of one of the millions of the audio samples in a song, and so on. Figure 4.28 in Chapter 4 explains this example in more detail.

Most of the common file formats fall under one of the following **data type** categories: text and hypertext, graphics, audio, animation and video.

Text and hypertext

Text is data that consists of strings of printable characters separated by white space and other non-printing characters. Hypertext is simply text that acts as a link to another document or another place in the same document.

Graphics

A graphic is a picture or diagram used to convey information easily to the user. There are many forms of graphics that can be used by the computer, and they can be incorporated into many types of applications. Graphics files fall into two main types: bitmaps or vectors. Bitmaps define the colour of each pixel on the screen, whereas vector graphics define an image using mathematical objects such as lines, squares and circles.

Audio

An audio file is digitised sound, which can be speech, music or sound effects. The amplitude and frequency of the sound are represented by a series of ones and zeros (Figure 3.17). As with graphics files, audio files can become very large, but there are compression techniques that can be used to reduce the size. Good-quality speakers or earphones are needed to listen to long audio files.

Animation

Animation is a series of graphic images that are shown rapidly in succession so that we see it as smooth motion. These graphics are mostly computer-generated or assembled and displayed using a software package. Morphing and warping images can be a lot of fun. Morphing software will change one image into another, whereas warping will create a distortion of a particular image.

Video

Video is now a common form of digital data. A video file, like animation, is also a series of graphic images. The images are commonly of the real world, such as people or events. Video files can have an audio element included in the same file. Video cameras and internet downloads are easily accessible. Video file sizes can be a problem, but streaming (continual transmission of audio and video files while being played) can be used to speed up download and playback.

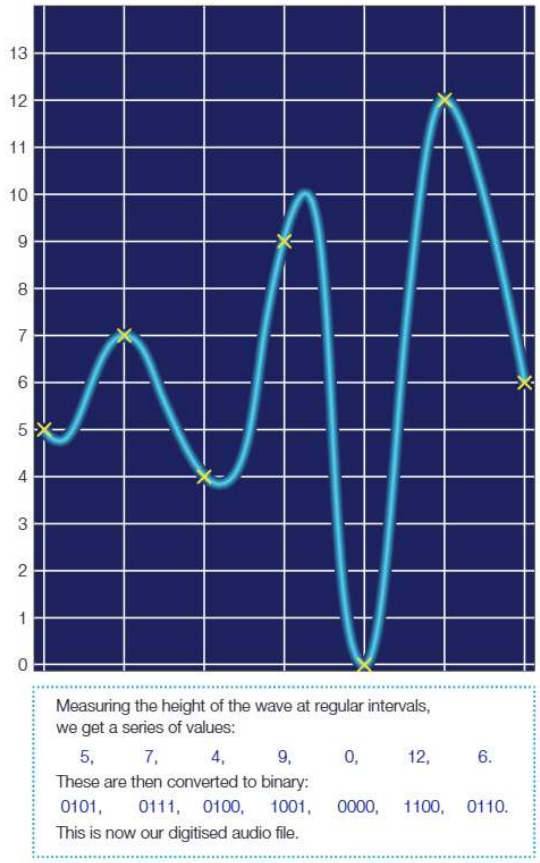
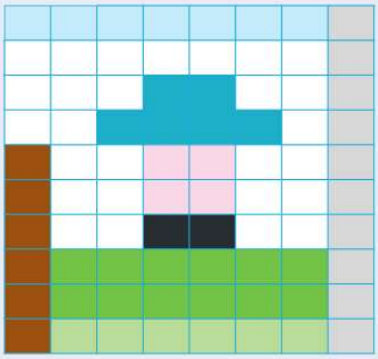


Figure 3.17 Converting analog sound to digital.

CASE STUDY

Understanding bitmap images

The amount of colour information in a picture is referred to as the colour depth or the number of bits per pixel. The greater the colour depth, the more colours that can be represented. The example shown here is using four bits to represent each colour, so it has a colour depth of four bits per pixel. Pixel refers to a picture element or one dot in the image.



Colour codes	
	1001
	0010
	1111
	1100
	0011
	1011
	0000
	1110
	0110

For a given picture size, more colours means larger file size.

- four bits per pixel (16 colours) appears as cartoon art
- eight bits per pixel (256 colours) is used in clip art and low-resolution pictures
- 24 bits per pixel (millions of colours) gives continuous tone such as you see in a photograph.

Questions

1. How many pixels are there in this graphic?
2. If it takes four bits to record the colour of a pixel, how many bits are needed to record the whole picture? What would be the file size for this graphic?
3. Write out the string of colour codes for each pixel across each of the 10 rows.
4. How many bits would be needed for colour depth of 24 bits per pixel?
5. What would be the file size if the resolution were increased to 100 pixels square at four bits per pixel? And at 24 bits per pixel?

Quantitative versus qualitative data

Data can be classified as either quantitative or qualitative.

Quantitative data is data that represents a 'quantity' or number of some form. Calculations can be easily performed on this data, as it is already in a numbered value to which a mathematical function can be applied. If collecting numbered ratings out of 10 for a new movie, a computer can easily calculate the average rating for the movie using a simple function.

Qualitative data does not have a distinct value or categorisation – for example, viewers' comments on a new film. This data may be equally, if not more, useful than quantitative data in understanding the quality of the film. However, a simple mathematical function cannot be applied to all of the comments in order to determine the overall impression of the film. Instead, we have relied on a human analysing all of the comments to get an overall understanding of the comments, including most common issues and praises.

Two significant areas in computer technology development are AI and natural language processing. These technologies can enable computers to understand natural language and therefore categorise and potentially quantify what is being said to allow for more efficient processing and analysis. This can lead to insights even a human may not capture.

When designing systems that collect data, it is important to consider whether quantitative or qualitative data will be the most beneficial, based on the information you aim to generate.

Common uses of data

We live in a world powered by data. In 2018, there were an estimated 33 zettabytes (33 trillion gigabytes) of data in the world, with 90 per cent of that data created in the past two years.

Data could simply be facts shared between people, such as event dates, instructions on how to make a cake or photos of your new cousin. It could also be specific numbers used to control automated systems, such as the sequencing and timing of traffic lights or the movement within a laser cutter. However, a lot of data has a more complex role and is processed and analysed to help make decisions and predictions.

Artificially intelligent systems such as OCR rely on many examples in order to 'learn' what a letter or number looks like. All these create data that allows the AI system to make a decision on the current character to be identified.

Predictions can be made about a real-world system using a model of that system. A model can be a function or equation that shows the relationship between two or more variables – for example, the average temperature over the months of the year. Modelling of systems relies on the analysis of large quantities of real data collected from the system. The greater the data used to build the model, the more accurate the model and the more confidence we can have in the predictions made based on the model.

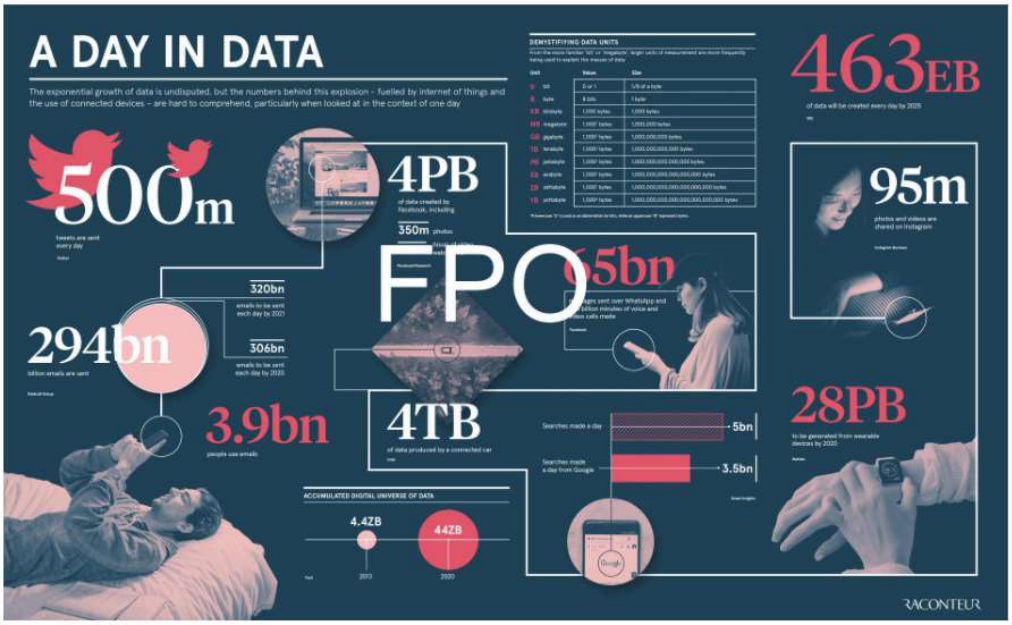


Figure 3.18 Infographic showing interesting facts about data use every day.

INFOBIT

YouTube: another petabyte every minute

Every day, YouTube needs a petabyte (PB) of new hard disk space to store the approximately 500 hours of video uploaded every minute. But what is a petabyte? It depends whether you are thinking in the decimal or binary system. The prefixes kilo, mega, peta and so on are terms from the decimal system, hence they define values based on the power of 10. Therefore, the decimal value $10^3 = 1000$ bytes or 1 kilobyte (KB). However, since computers work with just 0 and 1, the International Electrotechnical Commission (IEC) defines these values in the binary system with powers of 2. The binary value of $2^{10} = 1024$ bytes is called a kibibyte (KiB) to avoid confusion. So, the decimal term 'petabyte' is $10^{15} = 1\,000\,000\,000\,000\,000$ bytes, while a pebibyte (PiB) is $2^{50} = 1\,125\,899\,906\,842\,624$ bytes. When comparing KB and KiB, the difference is only 2.4 per cent, but when we compare PB and PiB, the difference is 12.6 per cent. So, as data storage requirements continue to increase, it's important to know the difference to make sure we are getting value for money!

Table 3.2 Decimal and binary values

Prefix	Symbol	Power of 10 (SI unit)	Equivalent power of 2 (IEC Unit)	Percentage difference IEC/SI
Kilo-	k or K	$10^3 = 1000$ bytes (1 KB)	$2^{10} = 1024$ bytes (1 KiB)	2.4%
Mega-	M	$10^6 = 1\,000\,000$ bytes (1 MB)	$2^{20} = 1\,048\,576$ bytes (1 MiB)	4.9%
Giga-	G	10^9	2^{30}	7.4%
Tera-	T	10^{12}	2^{40}	10%
Peta-	P	10^{15}	2^{50}	12.6%

ACTIVITIES: 3.4 DATA AND INFORMATION

1. Explain the difference between data and information. Provide an example to support your response.

.....

.....

.....

.....

.....

.....

.....

.....

2. Think of a project you completed last year. If you were to design a survey to get feedback on your project from your peers, what quantitative and qualitative data would you collect?

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

3.5 People and processes

There are two general categories of people involved in information systems: the users and the technical experts involved in the development, operation and management of the system. For the technical experts, the varied and complex nature of information systems means there are countless roles and responsibilities. Users are involved in the development, operation and management of the system. A career in information and software technology is obviously not for everyone, but for those interested in working in this area, there are many possibilities from which to choose. People with skills and knowledge working with information technology (IT) are highly sought after in many industries, including health care, education, defence, agriculture, engineering and entertainment.

Users

High-quality, successful information systems have been designed and developed with the users' needs always in mind. A system needs to have logical processes and an intuitive interface to make it user-friendly. The processes a user applies to the system should also be clearly outlined in documentation, such as a user manual.

Consider an information system you use regularly, such as an LMS like Google Classroom. Common processes you perform as a user include logging in, navigating, adding text, searching, displaying resources, uploading files, and so on. How easy is it to use the system? Are there any other processes you wish you could do in the system?

Developers

Computer scientists

Computer scientists design and develop hardware and software technologies. Specific tasks may include providing new approaches to security or devising improved methods for using computers in areas such as robotics, computer vision, medical research or digital forensics. Computer scientists generally require in-depth knowledge of the theoretical foundations of computer systems and the ability to construct and comprehend complex algorithms. Most professionals have tertiary-level computer engineering, software engineering or computer science qualifications.

Computer programmers

A programmer writes the code used to create, update and maintain software packages. A programmer also assists systems analysts and project managers to determine the type of software a user or client needs. The programmer translates the solution into a programming language and must also test and debug the code to ensure it is error-free. These professionals mostly have studied tertiary-level computer science.

Software engineers

Software engineers are involved in the design and development of software for operating systems and application programs. They use different programming languages, depending on the purpose of the program, and solve technical problems in existing software. A software engineer will have likely studied tertiary-level software engineering courses.

Systems analysts

Systems analysts play a critical role in the development and upgrading of information systems. The systems analyst's task is primarily to solve problems related to computer technology by designing and managing the development of information systems. Most analysts specialise in specific types of systems, such as accounting systems, financial systems or engineering systems. A systems analyst may have studied tertiary-level business IT courses in information systems.

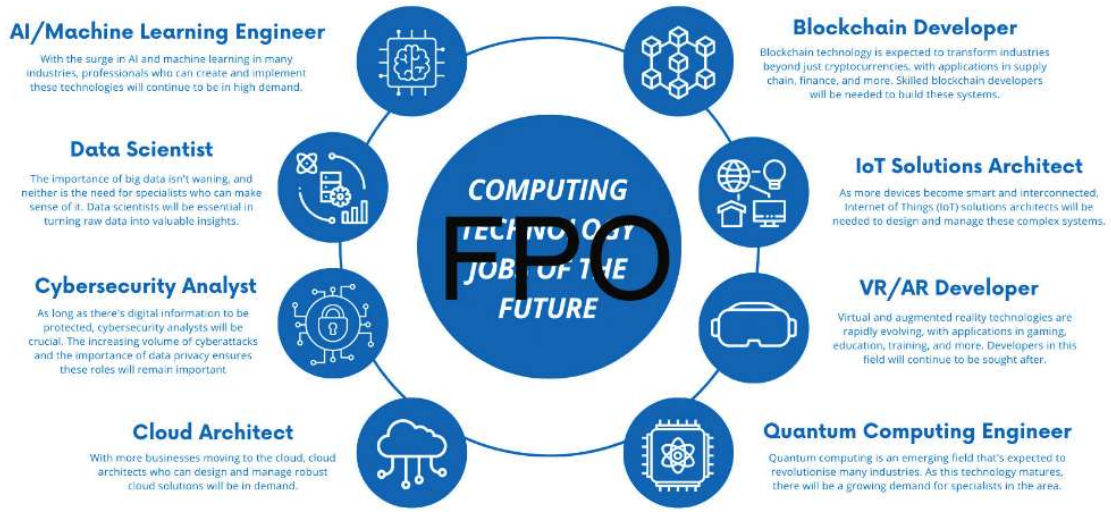


Figure 3.19 Computing technology future jobs.

Operators and administrators

Computer operator

The job of a computer operator is to keep powerful, essential computers, such as servers, operational. Responsibilities include keeping the software up to date, ensuring there is enough available storage and memory, checking the physical environment allows for healthy functioning of the computer, and identifying problems with the computers. The role of a computer operator also includes working with any specialised system software running on the computers.

Database administrator

Database administrators create and maintain databases used within an organisation. This may be on a small scale using database management systems or it may involve working with data warehouses. Databases are used across many processes within an organisation, so the administrator may be required to work with people such as systems analysts and programmers to help integrate database functions into new projects.

IT support staff

Support staff provide assistance for people who are having problems with their computer system. Most large organisations have support staff or help desks. A help desk is a department within an organisation that responds to users' technical questions. Support staff should be able to recognise and interpret the meaning of common error codes. They also need to be able to fix general problems associated with the daily use of hardware. More specialised hardware tasks, such as the installation, repair and maintenance of computers, servers and other devices, are mostly the responsibility of technicians.

Training specialists

The rapid spread of computers has placed more and more demands on people in terms of the technology skills they need to work, live and learn. Training specialists are educators who plan, coordinate and teach a range of activities related to information and software technology.

Management

Chief information officer

A chief information officer (CIO) is an increasingly common senior management role in many organisations, reflecting our increasing reliability on information systems. The CIO's role is to lead the planning and operation of information systems' function so that it aligns with the goals of the organisation. Key jobs include budgeting, scheduling and defining requirements for the information systems. Although knowledge and experience in an IT field is beneficial, it is more important for the CIO to understand the business and have good management skills, including effective communication.

Project managers

The principles of good project management apply across every industry and can be applied to many tasks, from making a dress to constructing an aeroplane. The field of information and software technology includes projects such as the creation of websites, video games, hardware devices, computer networks and operating systems. A **project manager** is required to keep the process on schedule and within budget by providing clear communication about goals, responsibilities, performance and expectations.

Information security officer

With nearly all organisations and their data connected to the internet, it is essential to have someone responsible for defining and implementing effective information security policies and protocols. The information security officer's role is to prevent external threats, such as viruses and hacking, and internal threats, such as the intentional or unintentional leaking of sensitive information. **Cybersecurity** is a rapidly growing field in the IT industry.

ACTIVITIES: 3.5 PEOPLE AND PROCESSES

1. Explain why you think users are important to consider when developing an information system.

.....

.....

.....

.....

.....

.....

.....

.....

2. Research a computing role of your choice.
a. Outline the educational requirements.

.....

.....

.....

.....

b. Describe key duties and responsibilities of the role.

.....

.....

.....

.....

.....

.....

.....

.....

3.6 Information systems: considerations and cybersecurity principles

One way to represent an information system is to show all the elements and the impacting environment using an 'information system in context' diagram. This diagram includes all of the elements covered so far in the chapter, wrapping it all up in the context of the environment in which the system will be implemented. No information system can be successfully designed and implemented unless everything that could impact the system and everything the system could impact has been considered and acted upon. Figure 3.20 is a template for creating a diagram for any system you are documenting.

Sociological considerations

When considering the social issues relating to the implementation of the system, it is important to consider both the positive and negative impacts it could have on individuals and the community.

For example, developers of the Opal travel card system would have considered the benefit to users of saving time and money by having an account and card that included multi-trip fare reductions and automatic top-up of credit. They would have also needed to consider the issue of people feeling their privacy was at risk and not wanting to have personal data being tracked and stored, so provided the option of still using a card without being linked to an account.

Information systems must also be designed with consideration of the perspectives of diverse groups including Aboriginal and Torres Strait Islander Peoples, culturally and linguistically diverse people, people of different ages and genders, and people with disabilities. For example, avoiding bias in data analysis, ensuring content is accessible to people with disabilities such as following World Wide Web Consortium (W3C) standards for machine-readable text, and respecting the Indigenous Cultural and Intellectual Property (ICIP) protocols relating to Aboriginal peoples' rights to their heritage, knowledge and cultural expressions.

INFOBIT

AI systems are commonly used to analyse medical imagery. In the area of chest radiography, it has been shown that models for pathology classification have a higher rate of underdiagnosis for underrepresented sub-populations due to sampling bias in the training data set. The use of these systems could therefore increase the probability of those patients being sent home without receiving the care they need.

Technical considerations

Technical considerations for an information system involve understanding how the system will integrate with existing hardware, software or networking and, if there are compatibility issues, how these will be resolved.

If data exported from one system needs to be imported into another system, the data needs to be in a suitable format that can be read by the new system.

It is also essential to consider other external digital systems to ensure protocols are followed to allow for this interaction and communication of data. For example, most organisations selling products online use an online payment system such as PayPal. Their website and payment system need to be able to accurately and securely communicate with PayPal.

Information system - <Title>
<Purpose>

Digital System

Components:

Hardware	Software	Networks

Key functions:

Data and information

People and processes
People are those that use the system or are involved more directly with the functioning of the system, such as a system's administrator.
Processes include anything related to the collection, organisation, editing, transmitting and receiving, analysing, storing and retrieving, and displaying of data and information.

Process	Person / People

Context / Environment

Social considerations	Technical considerations	Legal considerations
How will/does the system impact (positive or negative) individuals and the community?	*What existing hardware, software or networking will the system need to integrate with? Does the system need to interact with other external information systems?*	*Does the system abide by laws such as privacy and copyright? Does the system facilitate possible illegal action by users?*

Figure 3.20 A template for documenting the key features of an information system in context.

Cybersecurity principles

With most systems existing online, cybersecurity should be a key focus in information system development. The following are some of the key cybersecurity principles that organisations and individuals should follow to protect themselves against cyber threats:

- Confidentiality: Sensitive information should be protected from unauthorised access. This can be achieved through measures such as encryption, access controls, and data classification.
- Integrity: Data should be accurate and not tampered with or modified without authorisation. This can be achieved through measures such as data backups, version control and digital signatures.
- Availability: Authorised users should have access to information and systems when needed. This can be achieved through measures such as redundancy, disaster recovery planning and access controls.
- Authentication: Only users who are authorised can access information or systems. This can be achieved through measures such as passwords, biometrics and multi-factor authentication.
- Authorisation: Users have access to only the resources and information that they are authorised to access. This can be achieved through measures such as role-based access controls and user permissions.
- Accountability: Users should be responsible for their actions and that there is a clear record of who has accessed what information and when. This can be achieved through measures such as audit logs, activity monitoring and incident response planning.

Overall, these principles form the foundation of a strong cybersecurity strategy, and organisations and individuals should implement them to better protect themselves against cyber threats.

Legal considerations

Many IT companies employee specialist legal advisers to ensure their systems meet legal requirements. Not meeting the requirements can be very costly if an incidence highlights a company's lack of consideration. Two common laws to which information systems should abide are privacy and copyright. Many information systems store personal information about users. To ensure this information is kept secure, the information system should use established processes such as secure passwords and encryption of data.

Another legal consideration is whether the system could be used to facilitate illegal action. For example, an online alcohol supplier should not sell products to people under the age of 18. Social media systems even have some level of responsibility to consider and address the illegal act of cyber bullying that could occur through their system.

CASE STUDY

Electronic bike and scooter sharing



Shutterstock.com/Cascade Creatives

Figure 3.21 Lime-E bikes for hire.

A current transport trend spreading around the world is the sharing of electronic bikes and scooters. Companies such as Lime provide the bikes and scooters, along with the complex information system controlling their sharing and maintenance.

Users download the mobile application, sign up for an account, then for a small fee can unlock the scooter using a Quick Response (QR) code on their phone, with further charges based on the distance travelled.

Although this concept has many merits, including low-cost, efficient travel and environmental benefits such as lower carbon emissions, in some cities there have been many issues that have limited the success of the system. This is because, like all information systems, the people using the system and the context of where the system is being implemented have a huge impact on its success.

Chapter review quiz

One big issue relating to people is that users are not following the outlined processes correctly, including riding in locations that are not allowed, or parking the bike or scooter in places that are inappropriate. Brisbane trialled the e-scooters before fully implementing them. One procedure put in place in order to address past issues is users were required to upload a photo of how they parked their scooter, which was analysed to determine if it was satisfactory. If a user parked badly a few times, the company would suspend their account. In order to minimise this issue even further, the latest technology in development includes an indicator on the light-emitting diode (LED) screen on the scooter to tell the user if they have parked in the wrong place.

Legal issues are a key consideration relating to the context of the system's implementation. Many countries and cities have laws that make it illegal to ride electronic scooters in public areas either above a certain speed limit or at all. An automated feature used to address this is the use of a speed limiter. Companies have also gone down legal paths in some cities to have laws changed or exemptions allowed.

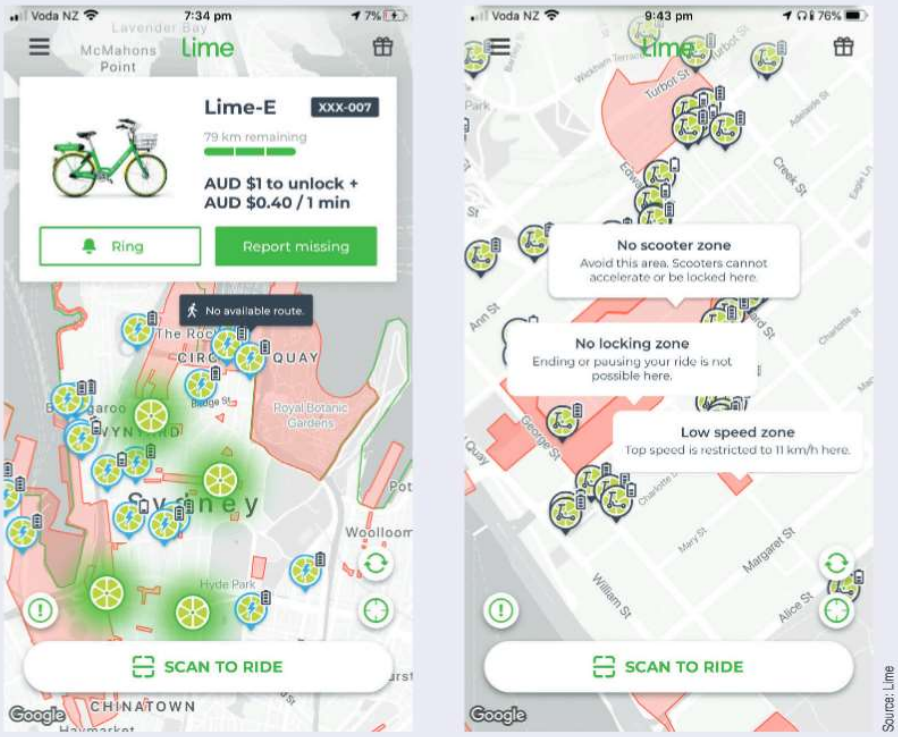


Figure 3.22 Screenshots from the Lime-E app showing key functionality of the map view, including locating available bikes or scooters and showing which areas have certain use restrictions.

Questions

1. Identify the purpose of this information system.
2. Outline some processes performed by users, highlighting any data or information involved.
3. This information system incorporates elements of a transaction processing system and a GIS. Describe the functionality in the elements relating to these categories of information systems.
4. Many information systems are trialled before they are fully implemented. Explain the benefits of trialling a system.
5. Summarise the considerations and issues associated with this system. Find a recent news article on how a system like this may have had a negative impact on its environment.

ACTIVITIES: 3.6 ISSUES AND CYBERSECURITY

1. Outline all cybersecurity principles that should be considered when collecting user data.

.....

.....

.....

.....

.....

.....

.....

.....

.....

2. Explain the difference between authentication and authorisation. Use an example of each to support your response.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

GLOSSARY OF TERMS

Application Software

.....

.....

.....

Authentication

.....

.....

.....

Authorisation

.....

.....

.....

Copyright

.....

.....

.....

CPU

.....

.....

.....

Cybersecurity

.....

.....

.....

Data

Data Analysis (Input, Process, Storage, Transmission, Output)

Data Types (Common)

Hardware

Information

.....

.....

.....

Networks

.....

.....

.....

Operating System

.....

.....

.....

Privacy

.....

.....

.....

Qualitative Data

.....

.....

.....

Quantitative Data

.....

.....

.....

Software

.....

.....

.....

Structured Data

.....

.....

.....

Unstructured Data

.....

.....

.....

User

.....

.....

.....